

GTA V: BLOODLINES — IMPLEMENTATION TOOLKIT

Complete Executable C# Source for Mission 01, Voice Batch Automation, Directory Staging & Testing Protocol

1. DEVELOPMENT SEQUENCE OVERVIEW

THE FOUR PRODUCTION ACTION STEPS

Moving from the design bible into active gameplay follows a strict chronological deployment sequence:

- **Action Step 1:** Staging GTA V root mod directories, installing Script Hook V .NET, and compiling BloodlinesCore.dll.
- **Action Step 2:** Executing the Python ElevenLabs batch synthesis script to generate all voice lines for the prologue.
- **Action Step 3:** Compiling and deploying the complete Bloodlines_M01.cs mission assembly (Vertical Slice).
- **Action Step 4:** In-engine live testing, coordinate verification, companion AI tethering audit, and checkpoint validation.

2. ACTION STEP 1: DIRECTORY STAGING & CONFIGURATION

GTA V MOD DIRECTORY TREE

```
Grand Theft Auto V/
├── ScriptHookV.dll
├── ScriptHookVDotNet.asi
├── ScriptHookVDotNet3.dll
├── scripts/
│   ├── BloodlinesCore.dll           <-- Shared runtime engine (compiled from Section 3)
│   ├── Bloodlines_M01.dll          <-- Mission 01 executable assembly
│   ├── BloodlinesConfig.ini        <-- Campaign configuration settings
│   └── Bloodlines/
│       └── audio/                  <-- Staged 44.1kHz 16-bit Mono WAV files
│           ├── M01_S1_01_ICE.wav
│           ├── M01_S1_02_GOHAN.wav
│           ├── M01_S1_03_GUESS.wav
│           └── ...
```

BLOODLINESCONFIG.INI SPECIFICATION

```
[General]
DebugMode = true
StartingMission = 1
EnableCompanionProtection = true
CompanionLeashDistance = 180.0

[Audio]
VoiceDirectory = scripts/Bloodlines/audio/
Volume = 1.0
DisplaySubtitles = true
SubtitleDurationMultiplier = 1.0

[Keybinds]
ReloadScripts = F9
SwitchToIce = NumPad1
SwitchToGohan = NumPad2
SwitchToGuess = NumPad3
TriggerSpecialAbility = Capital
SaveCheckpointDebug = Insert
RestoreCheckpointDebug = Delete
```

■ 3. ACTION STEP 2: PYTHON ELEVENLABS BATCH SYNTHESIS ENGINE

AUTOMATED VOICE INGESTION SCRIPT (BATCH_SYNTHESIZE_PROLOGUE.PY)

This standalone script iterates through the scripted dialogue lines for Mission 01, applies character-calibrated stability/similarity parameters, and outputs standard .wav files ready for the audio dispatcher:

```
import os
import requests

API_KEY = os.getenv("ELEVENLABS_API_KEY", "YOUR_ELEVENLABS_API_KEY")
TARGET_DIR = os.path.join("scripts", "Bloodlines", "audio")
os.makedirs(TARGET_DIR, exist_ok=True)

# Character Voice Profiles
VOICE_MAP = {
    "ICE": {
        "voice_id": "EXAVITQu4vr4xnSDxMaL", # Deep, resonant baritone
        "stability": 0.65,
        "similarity_boost": 0.85
    },
    "GOHAN": {
        "voice_id": "ErXwobaYiN019PkySvjV", # Crisp technical tenor
        "stability": 0.50,
        "similarity_boost": 0.80
    },
    "GUESS": {
        "voice_id": "VR6AewLTigWG4xS0ukaG", # High-energy street inflection
        "stability": 0.45,
        "similarity_boost": 0.75
    }
}

PROLOGUE_LINES = [
    ("M01_S1_01_ICE", "ICE", "Roost 4 confirmed. Two sentries on the upper deck. Target Mateo is in the stateroom sipping cognac. Moving into position."),
    ("M01_S1_02_GOHAN", "GOHAN", "Secondary bilge hatch corroded. Clamping RF jammer onto the hull sensor... grid is blind. Going inside."),
    ("M01_S1_03_GUESS", "GUESS", "Warehouse bay two is sweet. Biometric lock bypassed with physical tumblers. Pop... and we are in."),
    ("M01_S2_04_GOHAN", "GOHAN", "Alarm tripped! Stateroom safe is half-cloned! I got three shooters flanking the gangway!"),
    ("M01_S2_05_ICE", "ICE", "Wait... that voice. Devin?! Gohan?! What the hell are you doing on that boat?!"),
    ("M01_S2_06_GUESS", "GUESS", "Yo! Why am I hearing my high school graduation on my tactical radio?! Ice?! Gohan?!"),
    ("M01_S3_07_ICE", "ICE", "Mateo hit the slipway! He is on the armored launch! We blew the window!"),
    ("M01_S3_08_GUESS", "GUESS", "Get in! The entire port authority is rolling! Fifteen years and this is how we catch up?!"),
    ("M01_S3_09_GOHAN", "GOHAN", "Download aborted at forty-eight percent! The core recorded our biometrics—all three of us!")
]

def synthesize_prologue():
    for cue_id, speaker, text in PROLOGUE_LINES:
        profile = VOICE_MAP[speaker]
        out_path = os.path.join(TARGET_DIR, f"{cue_id}.wav")
        if os.path.exists(out_path):
            print(f"Skipping existing: {cue_id}.wav")
            continue

        url = f"https://api.elevenlabs.io/v1/text-to-speech/{profile['voice_id']}"
        headers = {"xi-api-key": API_KEY, "Content-Type": "application/json"}
        payload = {
            "text": text,
            "model_id": "eleven_monolingual_v1",
            "voice_settings": {"stability": profile["stability"], "similarity_boost": profile["similarity_boost"]}
        }
        res = requests.post(url, json=payload, headers=headers)
        if res.status_code == 200:
            with open(out_path, "wb") as f:
                f.write(res.content)
            print(f"Synthesized successfully: {cue_id}.wav")
        else:
            print(f"Failed {cue_id}: {res.status_code} - {res.text}")

if __name__ == "__main__":
    synthesize_prologue()
```

■ 4. ACTION STEP 3: COMPLETE EXECUTABLE C# MISSION SCRIPT (BLOODLINES_M01.CS)

VERTICAL SLICE MISSION ASSEMBLY (SCRIPTS/BLOODLINES_M01.CS)

This standalone Script Hook V .NET script controls Mission 01. It handles coordinate instantiation, AI relationship group hostility, real-time stage progression, dialogue triggers, checkpoint commits, and the Terminal Island bridge pursuit:

```

using System;
using System.Collections.Generic;
using System.Drawing;
using System.Windows.Forms;
using GTA;
using GTA.Math;
using GTA.Native;

public class Bloodlines_M01 : Script
{
    private enum MissionStage
    {
        NotStarted,
        Stage1_Infiltration,
        Stage2_FirefightAlarm,
        Stage3_ExtractionBreakout,
        Stage4_FreewayEscape,
        Completed,
        Failed
    }

    private MissionStage currentStage = MissionStage.NotStarted;

    // Prologue RAGE World Coordinates
    private static readonly Vector3 PosIceRoost = new Vector3(978.25f, -2988.42f, 42.15f);
    private static readonly Vector3 PosGohanBilge = new Vector3(962.11f, -3035.80f, -0.45f);
    private static readonly Vector3 PosGuessBay = new Vector3(1015.60f, -3002.15f, 5.90f);
    private static readonly Vector3 PosMateoBoat = new Vector3(920.45f, -3085.12f, 0.20f);
    private static readonly Vector3 PosSafeZone = new Vector3(720.50f, -2400.10f, 15.20f);

    private Vehicle getawayCar;
    private Vehicle mateoBoat;
    private Ped mateoPed;
    private List<Ped> cartelGuards = new List<Ped>();
    private Blip escapeBlip;

    private int missionTimer = 0;
    private bool stageTriggerA = false;
    private bool stageTriggerB = false;

    public Bloodlines_M01()
    {
        Tick += OnTick;
        KeyUp += OnKeyUp;
        Interval = 100; // 10 ticks per second
    }

    private void StartMission01()
    {
        Screen.FadeOut(800);
        Script.Wait(850);

        // 1. Initialize protagonists through BloodlinesCore
        BloodlinesSwitchManager.InitializeProtagonists(PosIceRoost, PosGohanBilge, PosGuessBay);

        // 2. Spawn Mission Entities
        Model t20Model = new Model(VehicleHash.T20);
        t20Model.Request(2000);
        if (t20Model.IsLoaded)
        {
            getawayCar = World.CreateVehicle(t20Model, PosGuessBay, 90f);
            getawayCar.CustomPrimaryColor = Color.Black;
            getawayCar.IsPersistent = true;
        }

        Model dinghyModel = new Model(VehicleHash.Tropic);
        dinghyModel.Request(2000);
        if (dinghyModel.IsLoaded)
        {
            mateoBoat = World.CreateVehicle(dinghyModel, PosMateoBoat, 225f);
            mateoBoat.IsPersistent = true;
        }

        // Spawn Cartel Guards
        SpawnCartelGuards();

        currentStage = MissionStage.Stage1_Infiltration;
        missionTimer = Game.GameTime;

        Screen.FadeIn(800);

        // Play Initial Voice Cue
    }
}

```

```

        BloodlinesAudioManager.PlayDialogue("M01_S1_01_ICE", "Ice", "Roost 4 confirmed. Two sentries on upper deck.
Moving into position.", 4000);
    }

    private void SpawnCartelGuards()
    {
        Model guardModel = new Model(PedHash.G_M_Y_MexGoon_01);
        guardModel.Request(2000);
        if (!guardModel.IsLoaded) return;

        int cartelRelGroup = World.AddRelationshipGroup("CARTEL_GUARD");
        World.SetRelationshipBetweenGroups(Relationship.Hate, cartelRelGroup, Game.Player.Character.RelationshipGroup);

        Vector3[] guardSpawns = new Vector3[]
        {
            new Vector3(968f, -3015f, 12f),
            new Vector3(960f, -3020f, 12f),
            new Vector3(980f, -3025f, 6f),
            new Vector3(995f, -3010f, 6f)
        };

        foreach (var pos in guardSpawns)
        {
            Ped guard = World.CreatePed(guardModel, pos);
            guard.RelationshipGroup = cartelRelGroup;
            guard.Weapons.Give(WeaponHash.AssaultRifle, 200, true, true);
            cartelGuards.Add(guard);
        }
    }

    private void OnTick(object sender, EventArgs e)
    {
        if (currentStage == MissionStage.NotStarted || currentStage == MissionStage.Completed) return;

        // Check Failure Conditions
        if (Game.Player.Character.IsDead || (getawayCar != null && !getawayCar.IsDriveable))
        {
            TriggerFailure("Getaway vehicle destroyed or operative neutralized.");
            return;
        }

        switch (currentStage)
        {
            case MissionStage.Stage1_Infiltration:
                if (Game.GameTime - missionTimer > 8000 && !stageTriggerA)
                {
                    stageTriggerA = true;
                    BloodlinesAudioManager.PlayDialogue("M01_S1_02_GOHAN", "Gohan", "Secondary bilge hatch corroded.
Clamping RF jammer... grid is blind.", 4000);
                }
                if (Game.GameTime - missionTimer > 15000 && !stageTriggerB)
                {
                    stageTriggerB = true;
                    BloodlinesAudioManager.PlayDialogue("M01_S1_03_GUESS", "Guess", "Warehouse bay two is sweet.
Biometric lock bypassed. Pop... we are in.", 4000);
                    Script.Wait(4000);
                    TransitionToStage2();
                }
                break;

            case MissionStage.Stage2_FirefightAlarm:
                // Check if all initial guards are suppressed
                int aliveGuards = cartelGuards.FindAll(p => p.IsAlive).Count;
                if (aliveGuards <= 1)
                {
                    TransitionToStage3();
                }
                break;

            case MissionStage.Stage3_ExtractionBreakout:
                // Check distance between player and getaway vehicle
                if (Game.Player.Character.Position.DistanceTo(getawayCar.Position) < 8f)
                {
                    TransitionToStage4();
                }
                break;

            case MissionStage.Stage4_FreewayEscape:
                if (Game.Player.Character.Position.DistanceTo(PosSafeZone) < 30f)
                {
                    TriggerSuccess();
                }
                break;
        }
    }

```

```

    }
}

private void TransitionToStage2()
{
    currentStage = MissionStage.Stage2_FirefightAlarm;
    BloodlinesCheckpointManager.CommitCheckpoint(1, 2, PosGuessBay, 90f, "T20");

    BloodlinesAudioManager.PlayDialogue("M01_S2_04_GOHAN", "Gohan", "Alarm tripped! Safe is half-cloned! Shooters on gangway!", 3500);
    Script.Wait(3500);
    BloodlinesAudioManager.PlayDialogue("M01_S2_05_ICE", "Ice", "Wait... Devin?! Gohan?! What the hell are you doing on that boat?!", 3500);
    Script.Wait(3500);
    BloodlinesAudioManager.PlayDialogue("M01_S2_06_GUESS", "Guess", "Yo! Why am I hearing my high school graduation on tac comms?! Ice?! Devin?!", 3500);

    // Alert Guards to attack Gohan and Ice
    foreach (var g in cartelGuards)
    {
        if (g.IsAlive) g.Task.FightAgainst(Game.Player.Character);
    }
}

private void TransitionToStage3()
{
    currentStage = MissionStage.Stage3_ExtractionBreakout;
    BloodlinesAudioManager.PlayDialogue("M01_S3_07_ICE", "Ice", "Mateo hit the slipway! He is on the launch! We blew the window!", 3500);
    Script.Wait(3500);
    BloodlinesAudioManager.PlayDialogue("M01_S3_08_GUESS", "Guess", "Get in! Port authority is rolling! Fifteen years and this is how we catch up?!", 3500);

    // Switch to Guess driving the getaway car
    BloodlinesSwitchManager.SwitchHero(BloodlinesSwitchManager.HeroID.Guess);
    Game.Player.Character.SetIntoVehicle(getawayCar, VehicleSeat.Driver);

    // Set companions into passenger seats
    if (BloodlinesSwitchManager.IcePed != null) BloodlinesSwitchManager.IcePed.SetIntoVehicle(getawayCar, VehicleSeat.Passenger);
    if (BloodlinesSwitchManager.GohanPed != null) BloodlinesSwitchManager.GohanPed.SetIntoVehicle(getawayCar, VehicleSeat.RightRear);
}

private void TransitionToStage4()
{
    currentStage = MissionStage.Stage4_FreewayEscape;
    BloodlinesAudioManager.PlayDialogue("M01_S3_09_GOHAN", "Gohan", "Download aborted at 48%! The core recorded our biometrics—all three of us!", 4000);

    escapeBlip = World.CreateBlip(PosSafeZone);
    escapeBlip.Color = BlipColor.Yellow;
    escapeBlip.ShowRoute = true;
    BloodlinesAudioManager.ShowSubtitle("~y~EVADE CHASE UNITS AND REACH CYPRESS DRAINAGE TUNNEL~s~", 5000);

    // Set Wanted Level to 3 Stars
    Game.Player.WantedLevel = 3;
}

private void TriggerSuccess()
{
    currentStage = MissionStage.Completed;
    Game.Player.WantedLevel = 0;
    if (escapeBlip != null && escapeBlip.Exists()) escapeBlip.Delete();

    Screen.FadeOut(1000);
    Script.Wait(1200);
    BloodlinesAudioManager.ShowSubtitle("~g~MISSION PASSED: GHOST IN THE DOCKYARD~s~", 5000);
    Screen.FadeIn(1000);
}

private void TriggerFailure(string reason)
{
    currentStage = MissionStage.Failed;
    BloodlinesAudioManager.ShowSubtitle($"~r~MISSION FAILED:~s~ {reason}", 5000);
    Script.Wait(3000);
    BloodlinesCheckpointManager.RestoreLatestCheckpoint(ref getawayCar);
    currentStage = MissionStage.Stage2_FirefightAlarm;
}

private void OnKeyUp(object sender, KeyEventArgs e)
{
    if (e.KeyCode == Keys.F11 && currentStage == MissionStage.NotStarted)

```

```
    {  
        StartMission01();  
    }  
}
```

■ 5. ACTION STEP 4: VERIFICATION, QUALITY ASSURANCE & DEPLOYMENT CHECKLIST

LIVE IN-ENGINE QA AUDIT PROTOCOL

AUDIT PHASE	TEST PARAMETER	PASS VERIFICATION CRITERIA
1. Model Hashes	Spawn verification for Ice, Gohan, and Guess.	Peds render correctly with FamCA_01, FamDNF_01, and BallaEast_01 without texture corruptions.
2. Audio Pipeline	ElevenLabs Mono WAV stream playback.	Audio plays cleanly from scripts/Bloodlines/audio/ with matching subtitle color codes (~r~, ~b~, ~g~).
3. Sky-Cam Switch	Native START_PLAYER_SWITCH execution.	Camera zooms into the sky and descends smoothly into the target ped without frame hitching or camera clipping.
4. Companion Leash	Distance tethering test (>180m separation).	Unselected companions automatically warp into empty vehicle seats or reposition to street sidewalks.
5. Checkpoint Engine	Forced player death on Stage 3.	Screen fades out, orphan peds/wreckage delete, and fresh T20 spawns with player at 100% armor.